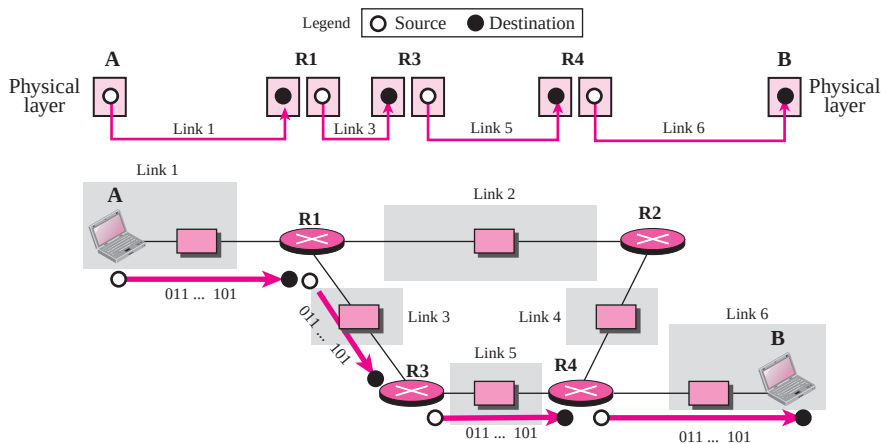


Figure 2.10 Communication at the physical layer

used by link 1; the layer connected to link 3 sends bits according to the format of the protocol used by link 3. It is the same situation with the other two routers involved in the communication.

The unit of communication at the physical layer is a bit.

The responsibility of the physical layer, in addition to delivery of bits, matches with what mentioned for the physical layer of the OSI model, but it mostly depends on the underlying technologies that provide links. We see in the next chapter that they are, for example, many protocols for the physical layer of LANs or WANs.

Data Link Layer

TCP/IP does not define any specific protocol for the data link layer either. It supports all of the standard and proprietary protocols. At this level, the communication is also between two hops or nodes. The unit of communication however, is a packet called a **frame**. A frame is a packet that encapsulates the data received from the network layer with an added header and sometimes a trailer. The head, among other communication information, includes the source and destination of frame. The destination address is needed to define the right recipient of the frame because many nodes may have been connected to the link. The source address is needed for possible response or acknowledgment as may be required by some protocols. Figure 2.11 shows the communication at the data link layer.

Note that the frame that is travelling between computer A and router R1 may be different from the one travelling between router R1 and R3. When the frame is received by router R1, this router passes the frame to the data link layer protocol shown at the left. The frame is opened, the data are removed. The data are then passed to the data